Prince Yadav

AI Engineer · LLM & RAG Systems

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B.Tech CSE (AI & ML), 2026, with two internships (IBM, Visys) and two projects live on the web: a RAG document Q&A system and a ship-inspection vision app. Python and C++, with DSA and OOP fundamentals. Based in Bengaluru; can join immediately.

TECHNICAL SKILLS

Languages	Python · C++ · SQL
CS Fundamentals	Data Structures & Algorithms · Object-Oriented Programming
AI / ML	PyTorch · Hugging Face · LoRA / QLoRA · RAG · OpenCV · SAM · LightGlue · scikit-learn · Time-series forecasting
Backend & Tools	FastAPI · Docker · Git · MLflow · Pinecone · ChromaDB · Supabase · Gemini, OpenAI & Groq APIs

EXPERIENCE

Research Intern — Visys · with a Manchester (UK) researcher live demo

Jul 2026 – Present

- Built ARMS, which finds where a ship image was taken from by matching it to the ship's 3-D CAD model (SuperPoint + LightGlue over 80 reference views, PnP pose solving): 12 of 12 test renders from new viewpoints located, typical error 0.16 m.
- Flags missing or moved parts with SAM 2.1 outlines: caught 29 of 30 removed parts with 2 false alarms in 153 checks. A verification gate rejects images it cannot trust: it correctly rejected all 10 photos of other ships. Live on Hugging Face GPU.

AI / ML Intern — IBM · ExpensexchangeAI project

Jun – Jul 2025

- Built ETL pipelines and forecasting models for a stock-portfolio analyser (20+ tickers): 76% directional accuracy, R^2 0.82.
- Cut forecast error 7–10% with Optuna tuning, Huber loss and feature engineering; tracked experiments and model versions in MLflow.

PROJECTS

RAG Lab — Document Q&A System GitHub · live demo

Jan 2026 – Present

- Hybrid retrieval over text, tables and figures: Pinecone dense search plus Supabase full-text / BM25 search, merged with Reciprocal Rank Fusion and reranked by a CrossEncoder; answers cite the source page.
- LLM-as-a-judge evaluation suite (groundedness, context and answer relevance): 0.95 / 0.90 / 0.95 across 80+ test questions.
- Cut server memory from over 1 GB to under 100 MB by moving embeddings to the Gemini API, so the app runs on a 512 MB free tier. FastAPI, Next.js, Docker.

MELD Emotion — Multimodal Emotion Recognition GitHub

Jan – May 2026

- Fine-tuned Llama-3.2-3B with LoRA as the text model, on 9,816 MELD TV-dialogue clips across 7 emotions.
- Trained a CNN-BiLSTM on Mel-spectrograms for audio and a 3D ResNet on video frames.
- Fused all three with TA-AVN cross-modal attention and Squeeze-Excitation gating; trained on an HPC GPU cluster.

Echo-D — Ultrasound Diagnostic AI GitHub

Aug – Dec 2025

- Offline pipeline to detect placenta accreta in ultrasound: SAM zero-shot segmentation and BLIP-2 radiologist-style captions.
- +14% accuracy over a U-Net baseline on 500+ anonymised scans (Dice, IoU, AUC, F1); Grad-CAM++ shows what drove each result.
- Built a synthetic ultrasound generator to offset scarce medical data; shipped as a Docker microservice with a REST API.

EDUCATION & ACHIEVEMENTS

B.Tech, Computer Science & Engineering (AI & ML) — UPES, Dehradun · CGPA 7.26 / 10

Aug 2022 – Jul 2026

Coursework: Data Structures & Algorithms · Machine Learning · LLMs & Generative AI · Computer Vision · Explainable AI

- 2nd Runner-Up, IBM Hackathon: designed and built an end-to-end AI solution under competitive evaluation.
- UPES DSW Ambassador (Jan 2024 – Feb 2025): student–administration liaison; ran AI seminars and mentored juniors.